mollie

Increase observability, not costs

Controlling telemetry data ingestion at scale



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Mollie on a page

VISION

Why we are on earth, our long term aspiration.

Eliminate financial bureaucracy

MISSION

Our purpose, our core activities and audience

Make money management effortless for every business in Europe

VALUES

What we value, what drives us, our DNA

Be loved
Be bold
Be authentic

PRINCIPLES

How we behave, how we show up, what we hold each other accountable to

Make our customers proud

Get things done today

Embrace change

Bold ownership, humble execution

Diverse and united

Speak up

Agenda

1. Problem Contextualization
2. Implementation Journey
3. Value & Impact
4. Key Learnings
5. QA

2004

Mollie was founded

2019

First funding round

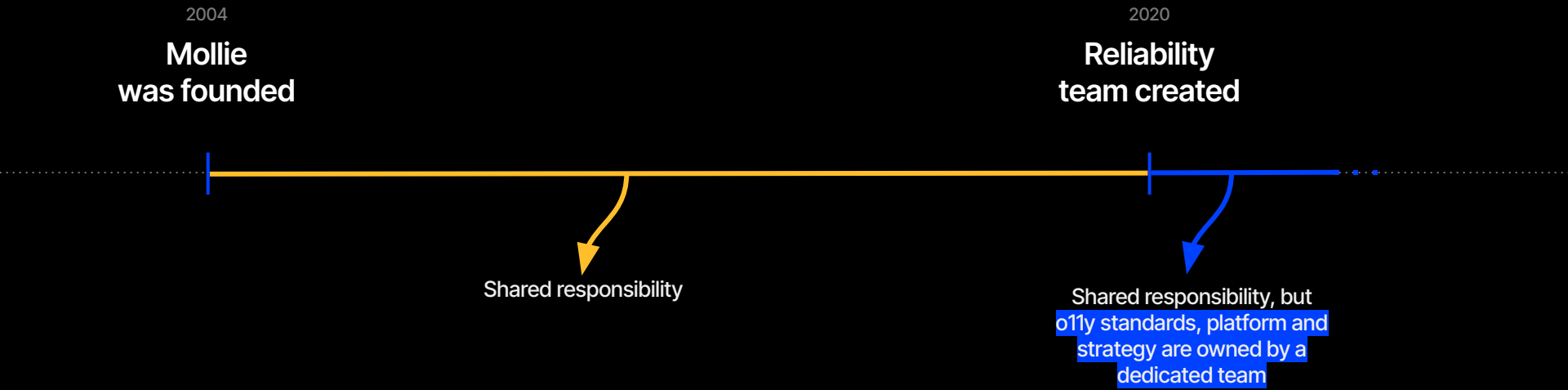
Start of growth phase



Recent & Rapid growth



Shared responsibility, no o11y standards, costs were rising


2004
Mollie was founded
2020
Reliability team created
Shared responsibility
Shared responsibility, but o11y standards, platform and strategy are owned by a dedicated team

Simple, right?



Educate teams



Create standards



Fix the sources

Simple, right? At the beginning, not really.



Small Central Team

Team size made it hard to coordinate with other teams. Adoption of new solutions was slow and complex.



PHP monolith

Various critical systems share the same codebase. Deployments are very frequent, resulting in large deployment trains.
For historical reasons used a metrics logger, which made attribution hard.



Shared infra module

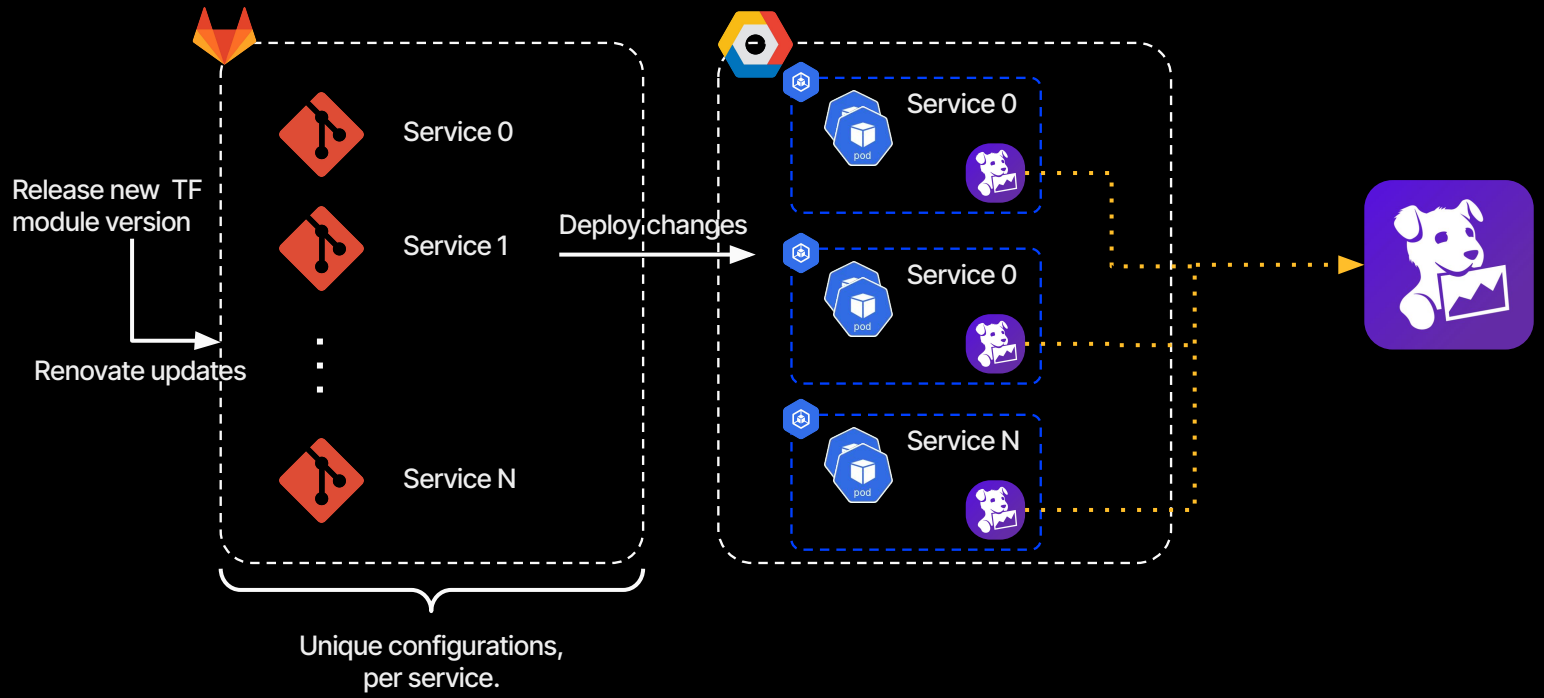
Central terraform module, makes it simple for application teams to provision infrastructure, but hinders the velocity and ability to roll out isolated changes.



Frustration Engineering

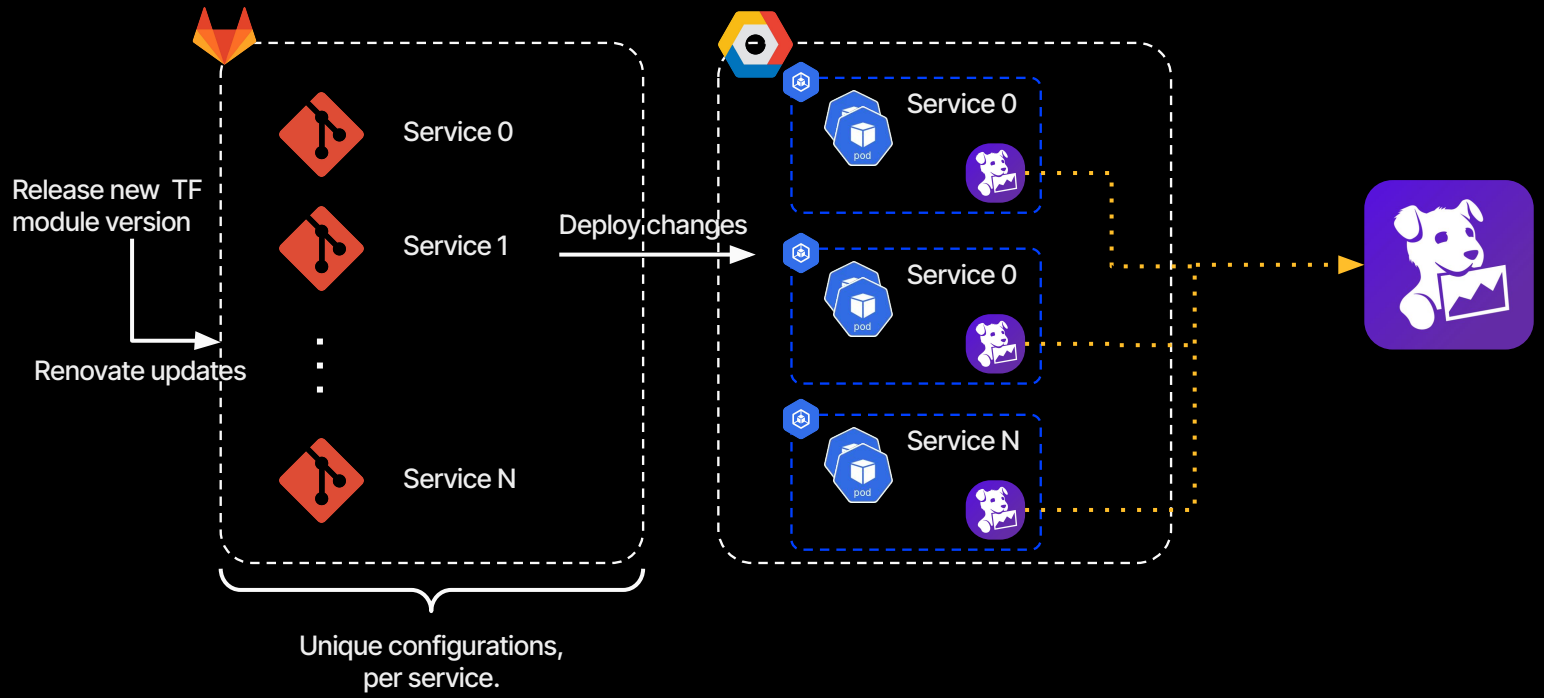
History of ambitious solutions, that lacked adoption. People came in, tried to solve the problem, and quit because they could not change what other teams were doing.

How were o11y infrastructure changes rolled out?



How were o11y infrastructure changes rolled out?

~ days/weeks



The problem



Rising o11y costs

- Low adoption of standards.
- Lack of attribution.
- External team dependencies.
- Team capacity.
- Vendor outages & expensive incidents.



Development Velocity

- Application engineers required velocity.
- Focus on feature development.
- Cost was not their problem.
- Systems had some observability.
- Teams were growing, fast.

Solution



Fire & Forget



No external dependencies



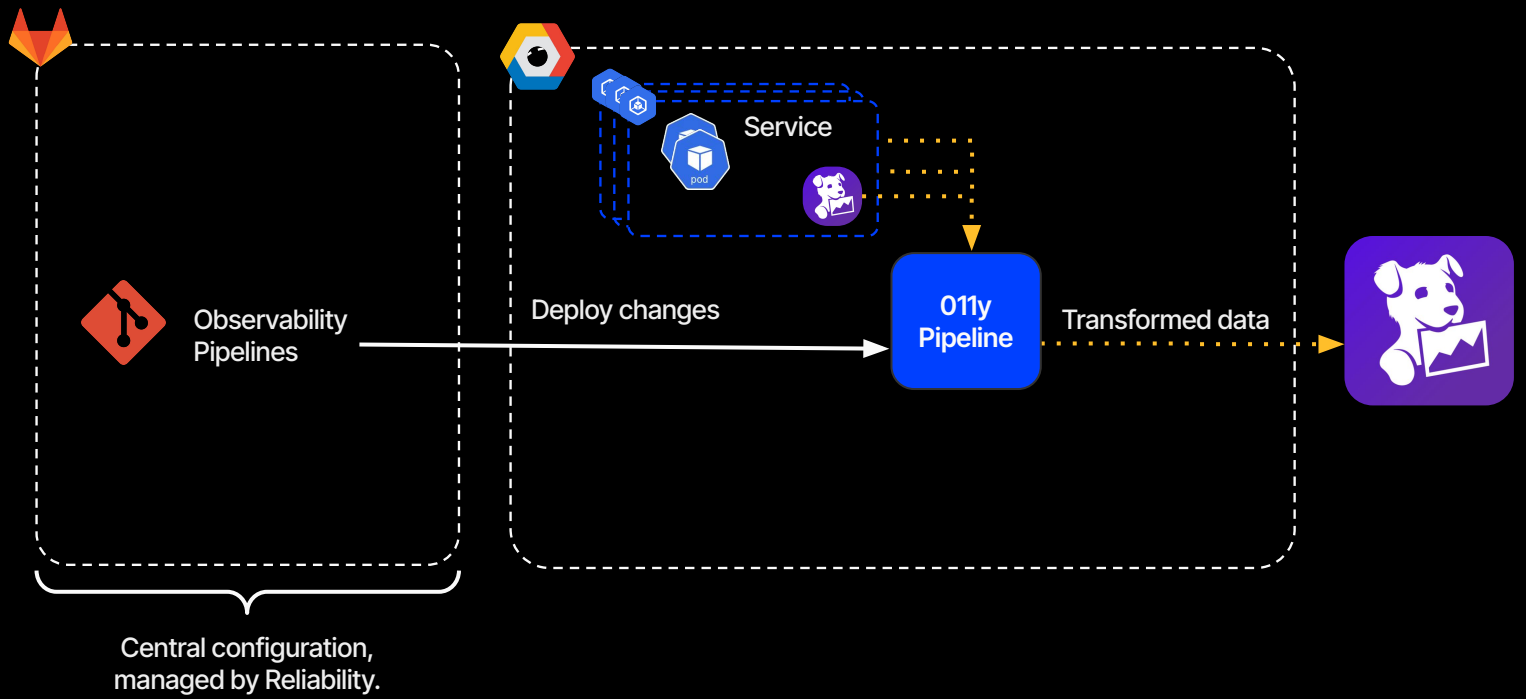
Minimal migration efforts



Data transformations

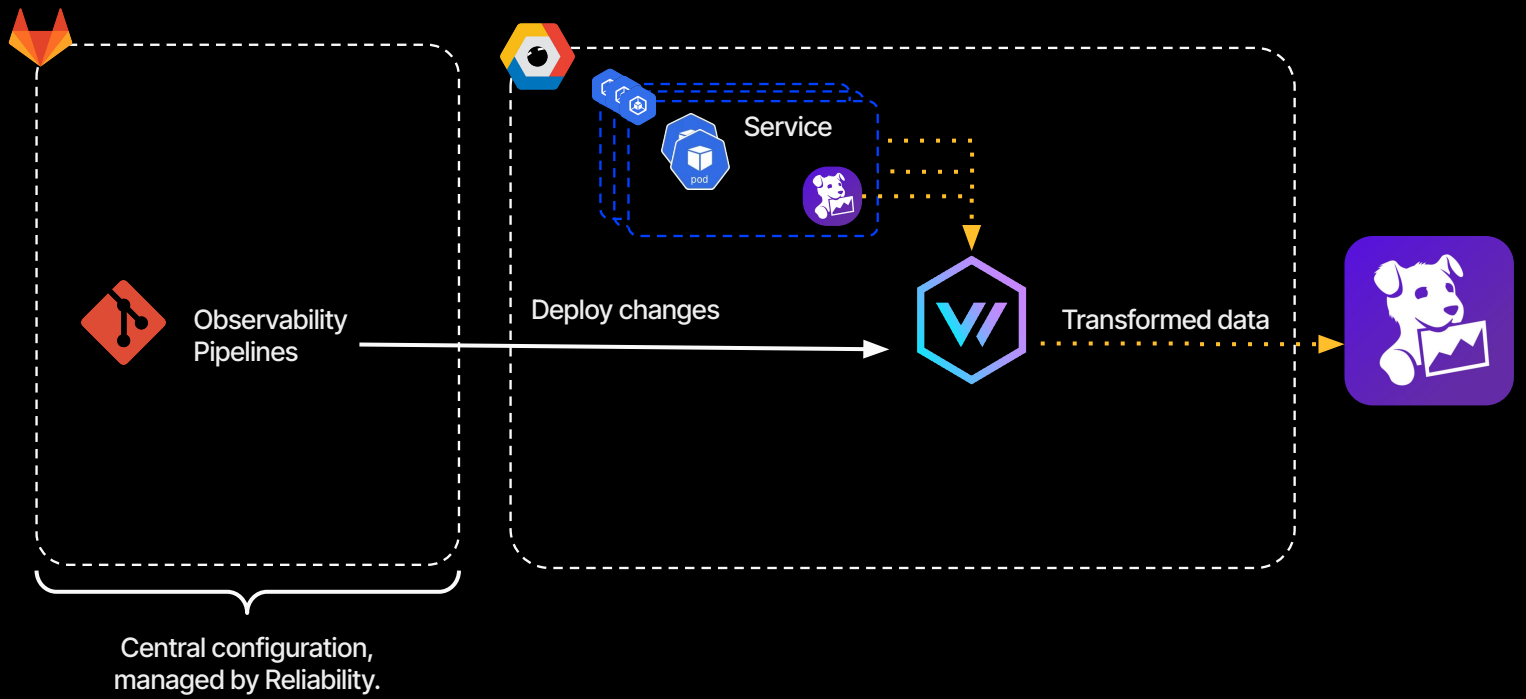
How are o11y infrastructure changes rolled out?

~ minutes



How are o11y infrastructure changes rolled out?

~ minutes



Why Vector?

Performance 🚀

High-performance, memory-efficient, and optimized for streaming ingestion.

Unlike OTEL Collectors' poll-based approach, Vector's event-driven model reduces resource usage and latency bottlenecks.

Plug and Play 🔌

Seamless integration with sources/sinks and existing telemetry pipelines.

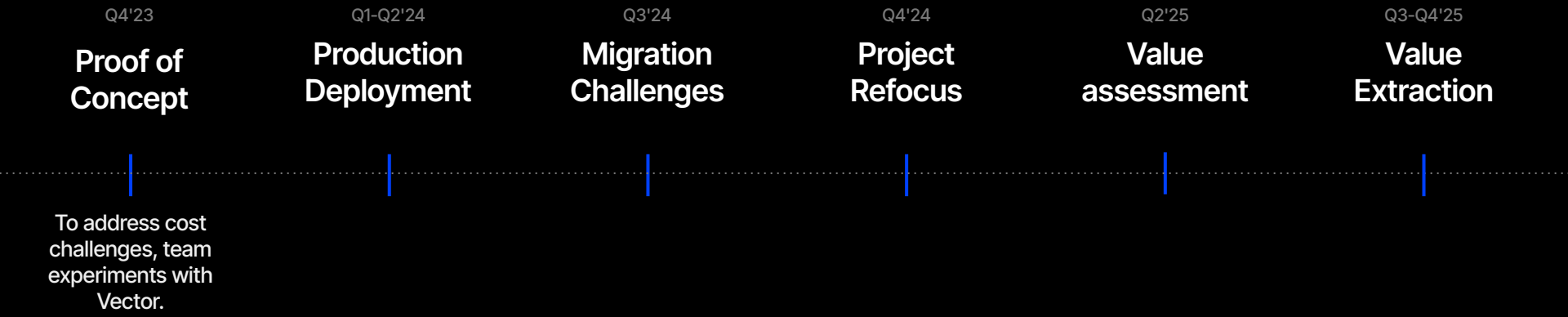
Minimal configuration changes compared to OTEL Collectors, which required significant structural adjustments.

Features ⚙️

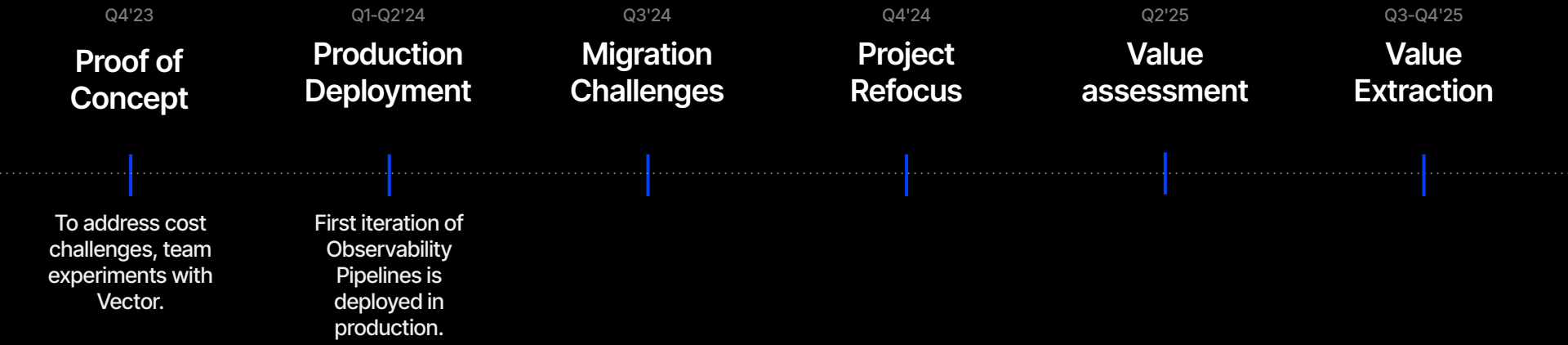
Built-in sources, sinks, and transformation capabilities for efficient data routing.

Programmable transforms allows the team to apply filtering and aggregation rules only once, affecting all projects.

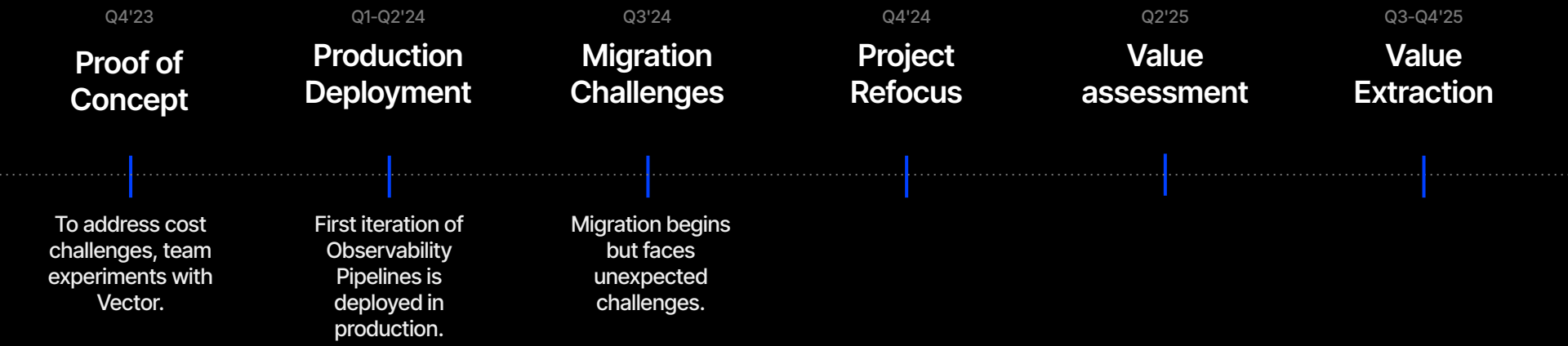
Implementation Journey



Implementation Journey



Implementation Journey

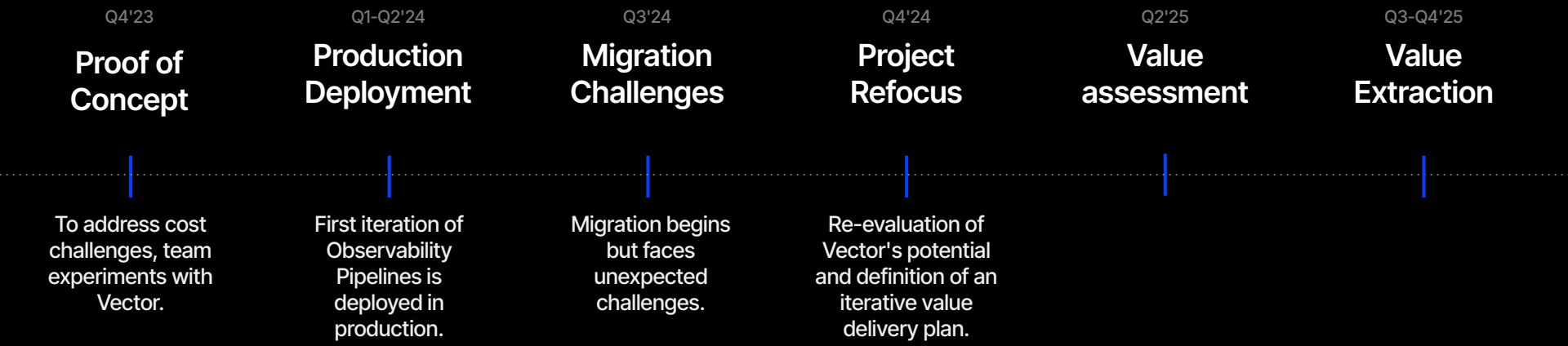


Implementation Journey

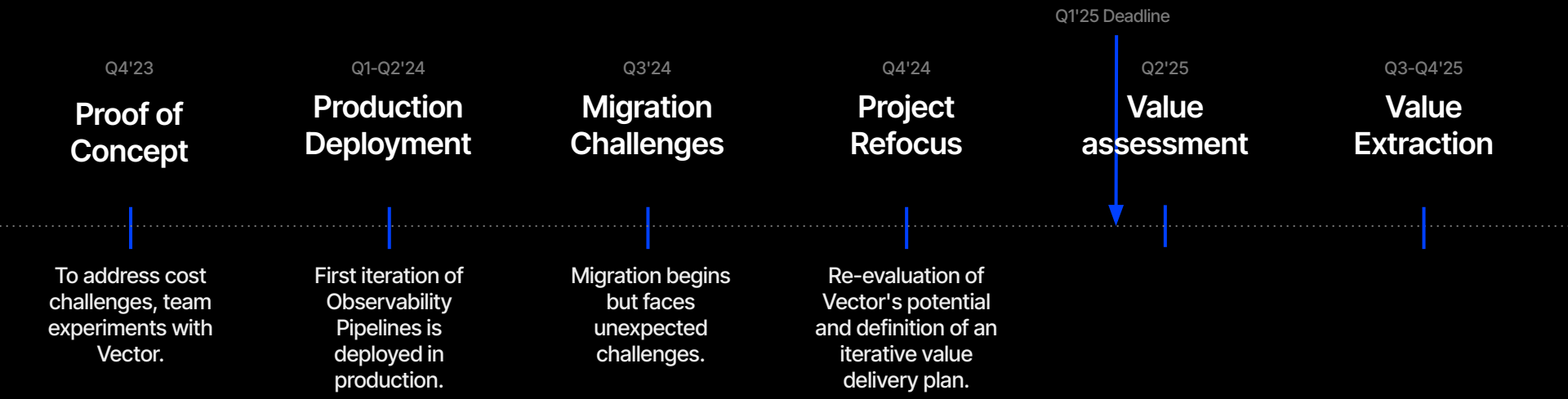


- No formal value analysis was conducted beforehand!
- Complex migration, faced issues to decouple o11y terraform module.
- Outages on critical services.

Implementation Journey



Implementation Journey



Implementation Journey

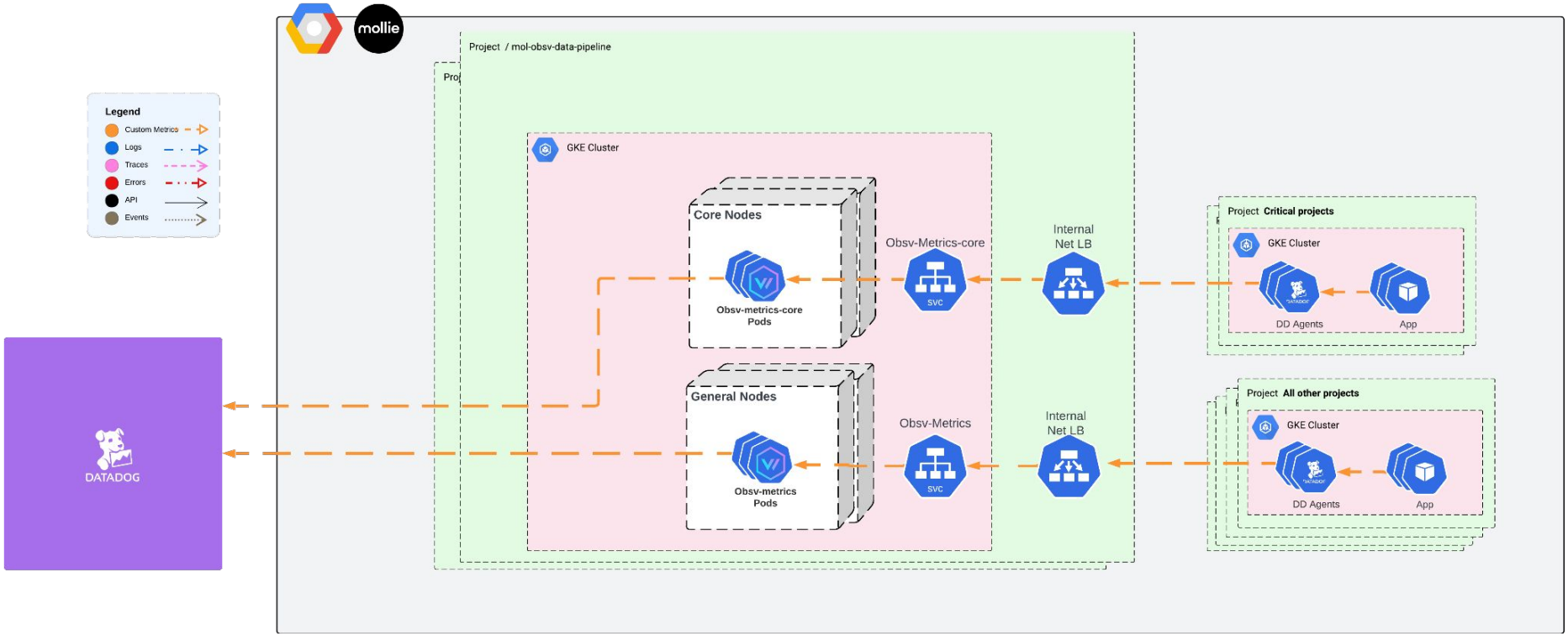


- Separation of services by criticality (core/non-core).
- Investigation of issues, in collaboration with strategic projects (e.g. dropped events, source lag...).
- Investment in monitoring and capturing data insights.

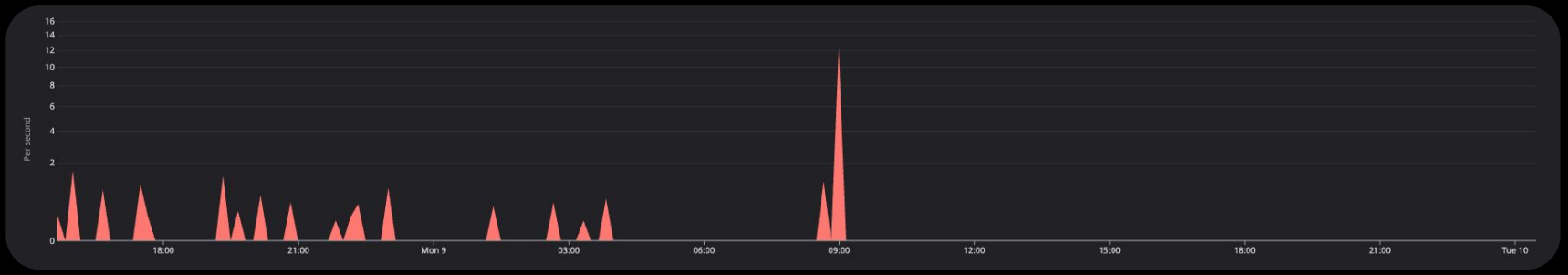
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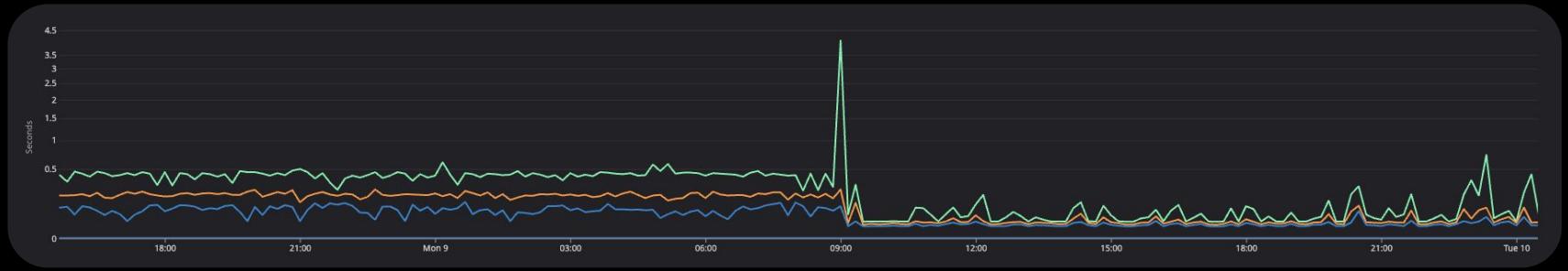
- Fully migrated o11y Terraform module out of general module.
- Improved production readiness and roll-out procedures.
- Phased onboarding of projects.


Legend
Custom Metrics
Logs
Traces
Errors
API
Events
mollie
Project / mol-obsv-data-pipeline
GKE Cluster
Core Nodes
Obsv-metrics-core Pods
Obsv-Metrics-core
SVC
Internal Net LB
General Nodes
Obsv-metrics Pods
Obsv-Metrics
SVC
Internal Net LB
Project Critical projects
GKE Cluster
DD Agents
App
Project All other projects
GKE Cluster
DD Agents
App
DATADOG

Investigation 1: Source Lag



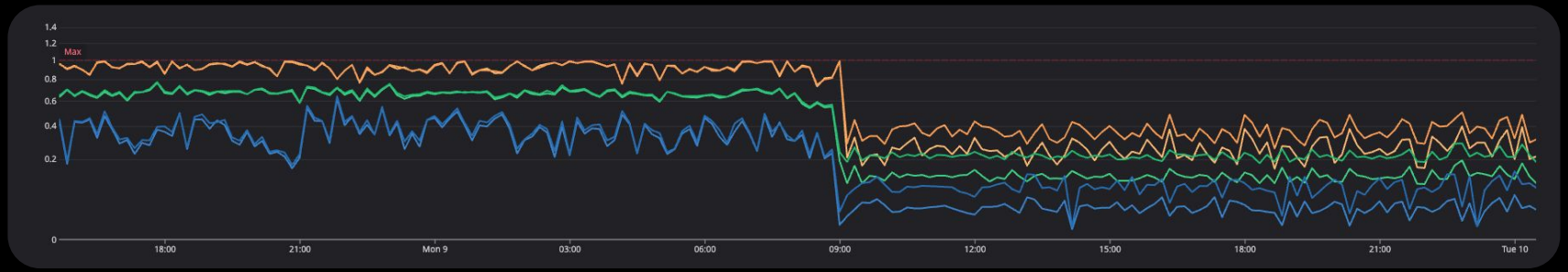
Dropped Events



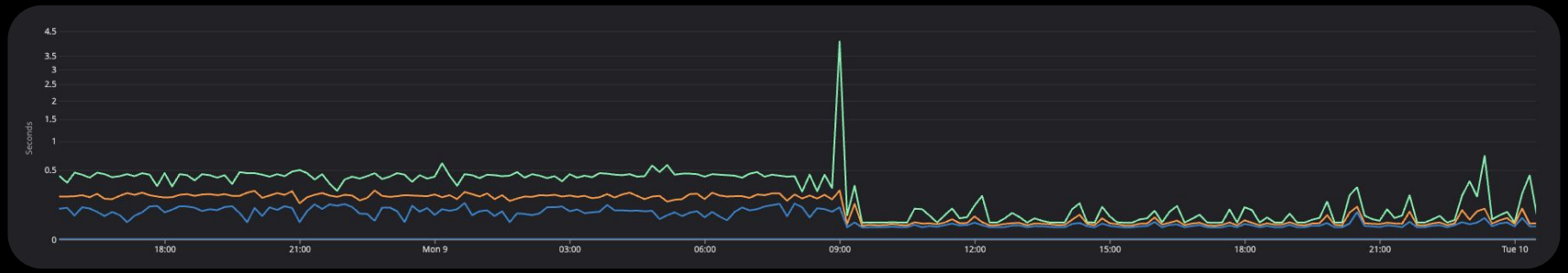
Source Lag

Investigation 1: Source Lag

P99
P95
P90

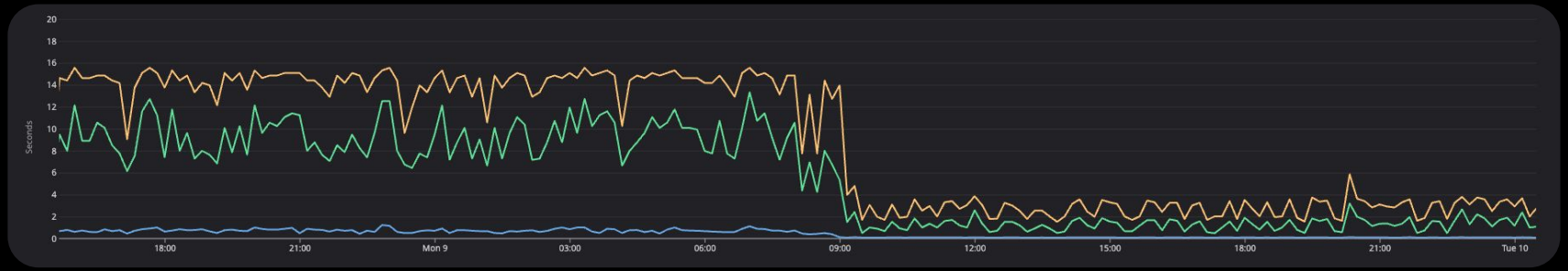


Utilisation

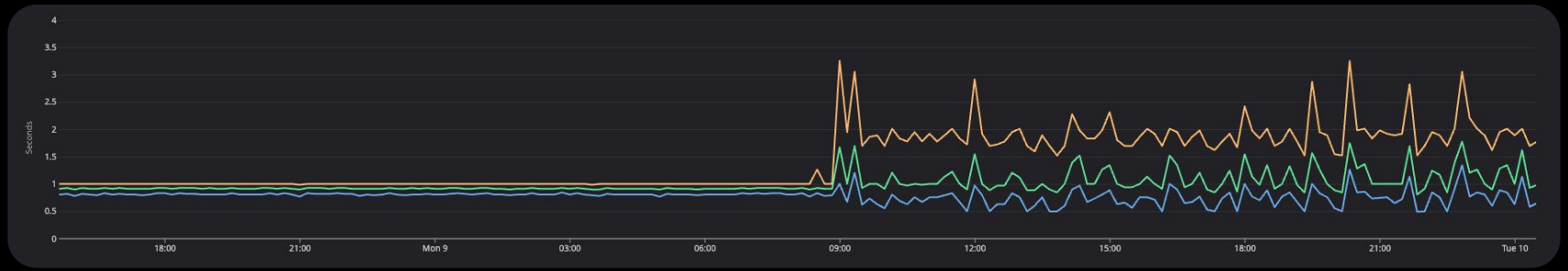


Source Lag

Investigation 1: Source Lag



Request Handling Duration



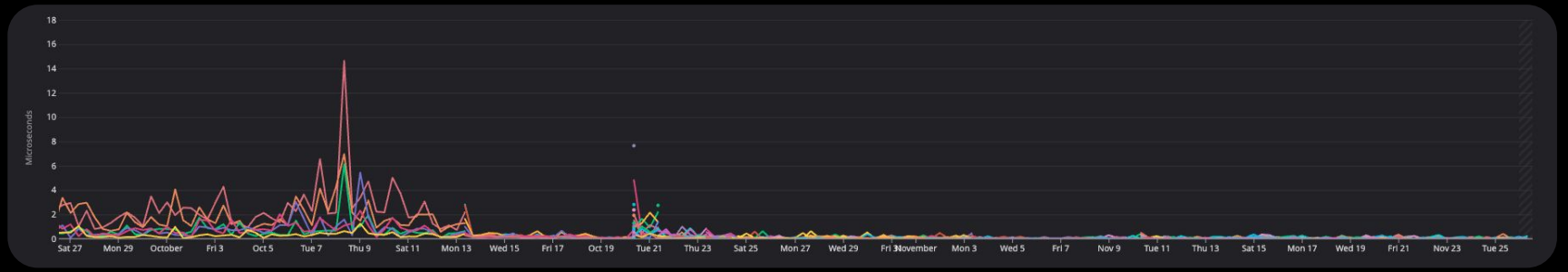
Client RTT

Investigation 2: CPU Throttling

P99
P95
P90

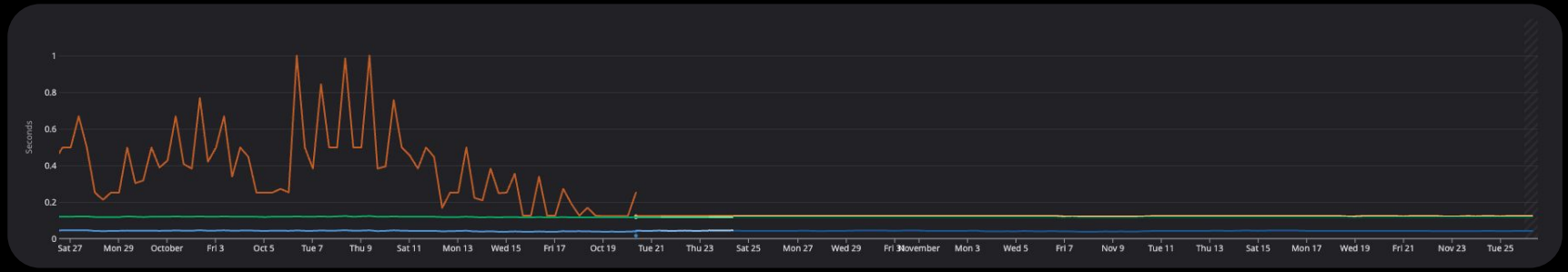


Dropped events

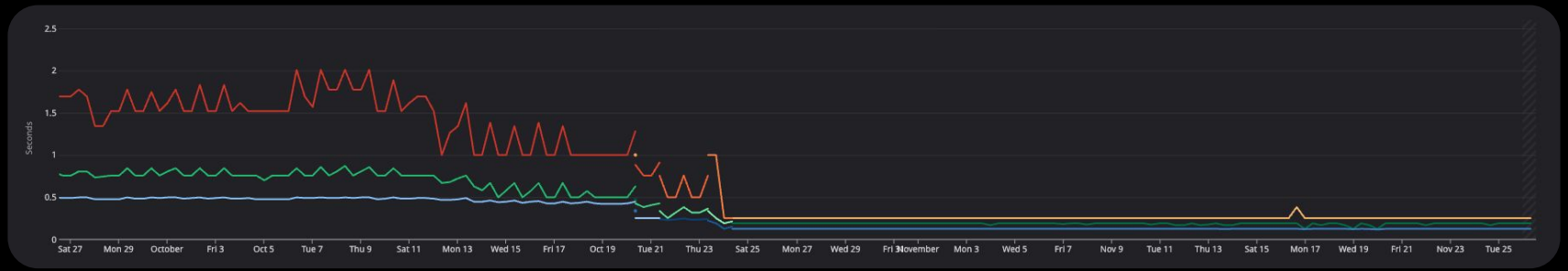


CPU Throttling

Investigation 2: CPU Throttling



Request Handling Duration



Client RTT

Implementation Journey

Q4'23

Proof of Concept

To address cost challenges, team experiments with Vector.

Q1-Q2'24

Production Deployment

First iteration of Observability Pipelines is deployed in production.

Q3'24

Migration Challenges

Migration begins but faces unexpected challenges.

Q4'24

Project Refocus

Re-evaluation of Vector's potential and definition of an iterative value delivery plan.

Q2'25

Value assessment

With data-backed insights, Vector’s trade-offs, tangible and intangible benefits are fully understood.

Q3-Q4'25

Value Extraction

Implementation Journey

Q4'23

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Q3-Q4'25

Value Extraction

All services onboarded. Start of experiments and value extraction phase.

First experiment: filtering unused metrics



High volume of unused metrics



Costly & hinders UI experience

First experiment: filtering unused metrics



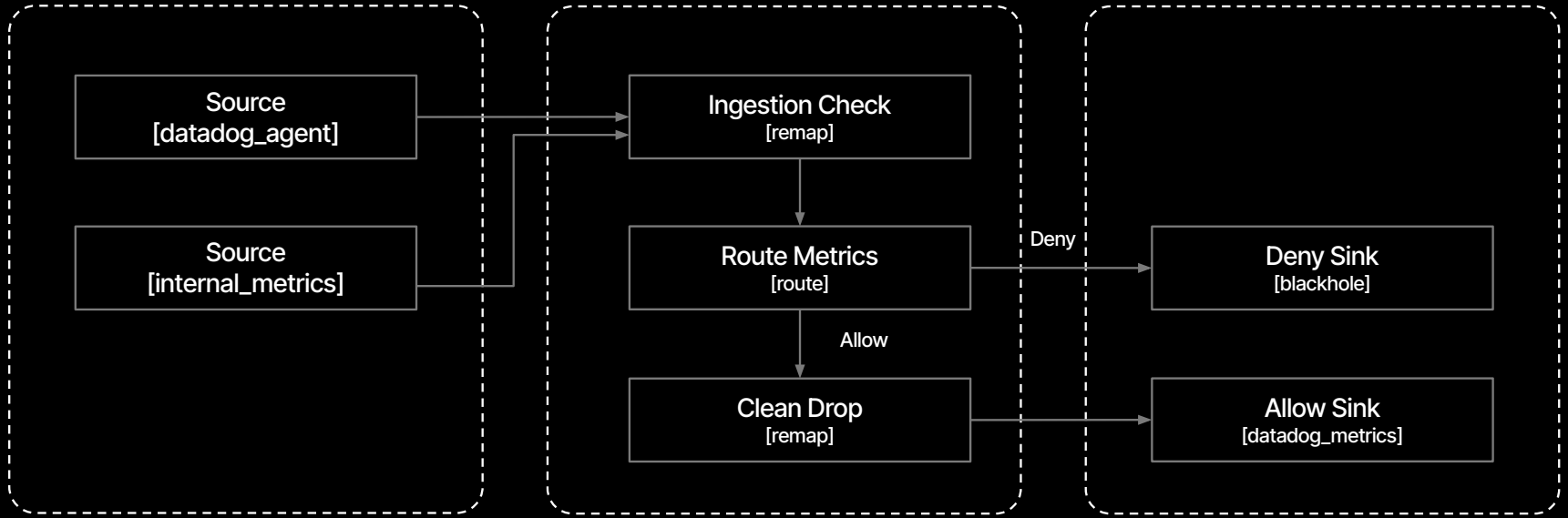
High volume of unused metrics



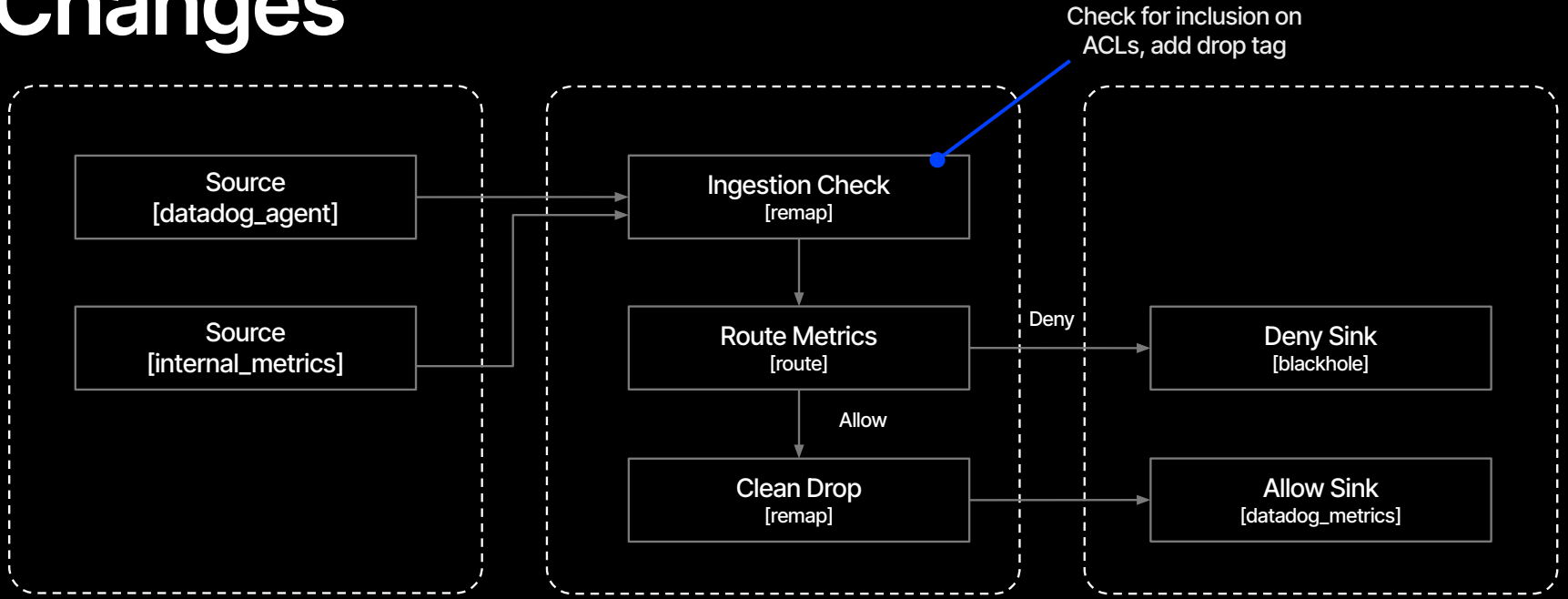
Costly & hinders UI experience

Around ~250k metrics that had not been queried for 30 days, nor used in any asset!

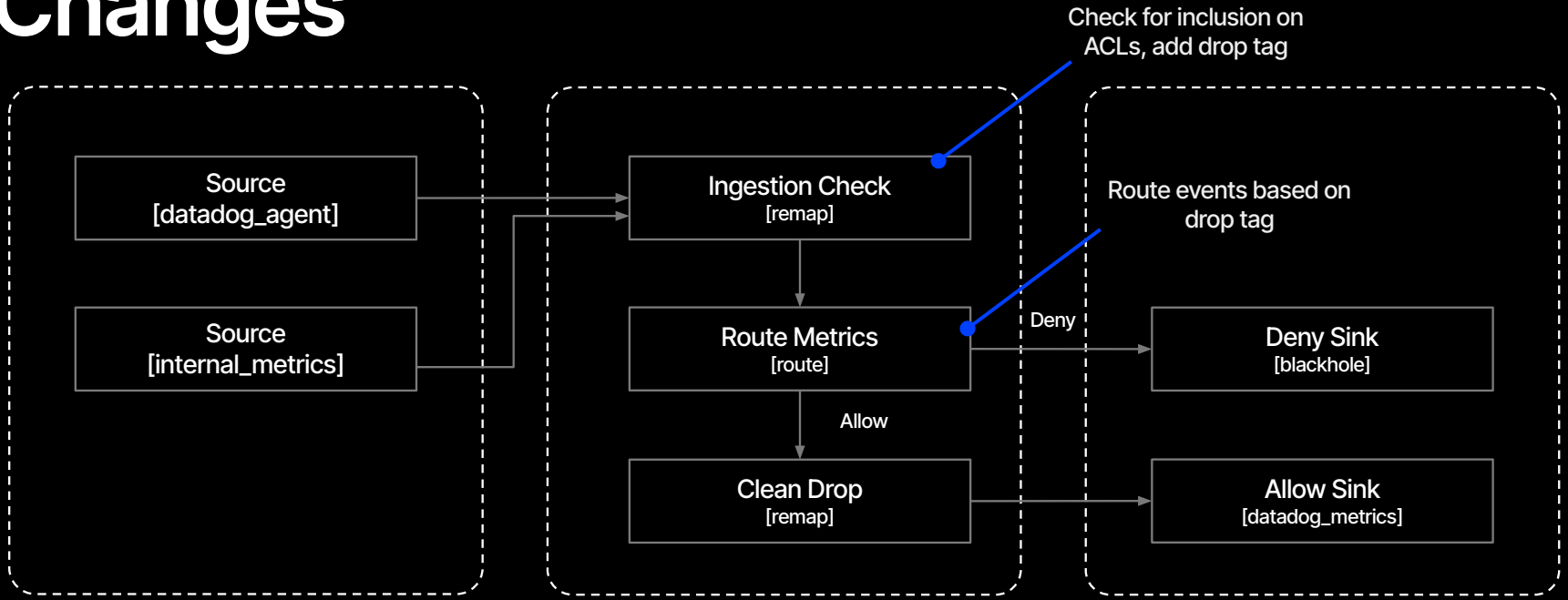
Vector Remap Language (VRL) Changes



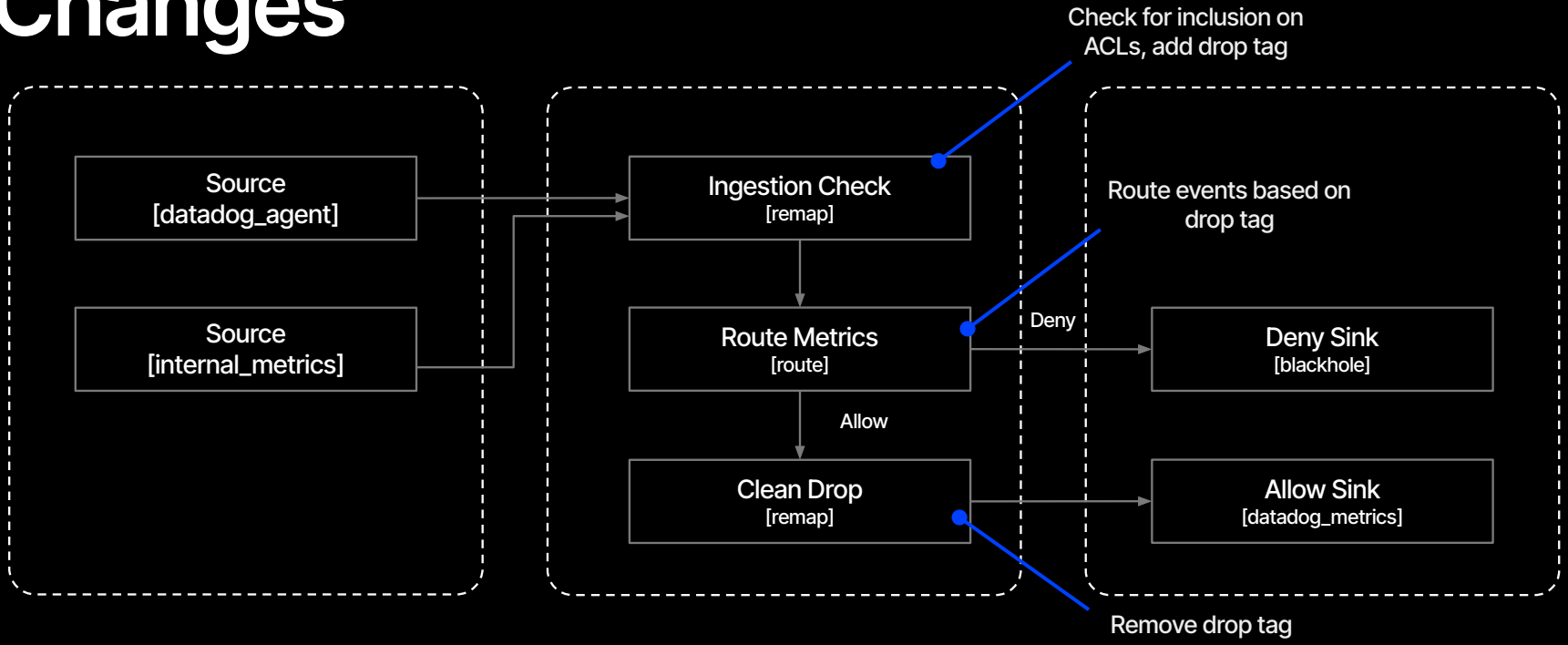
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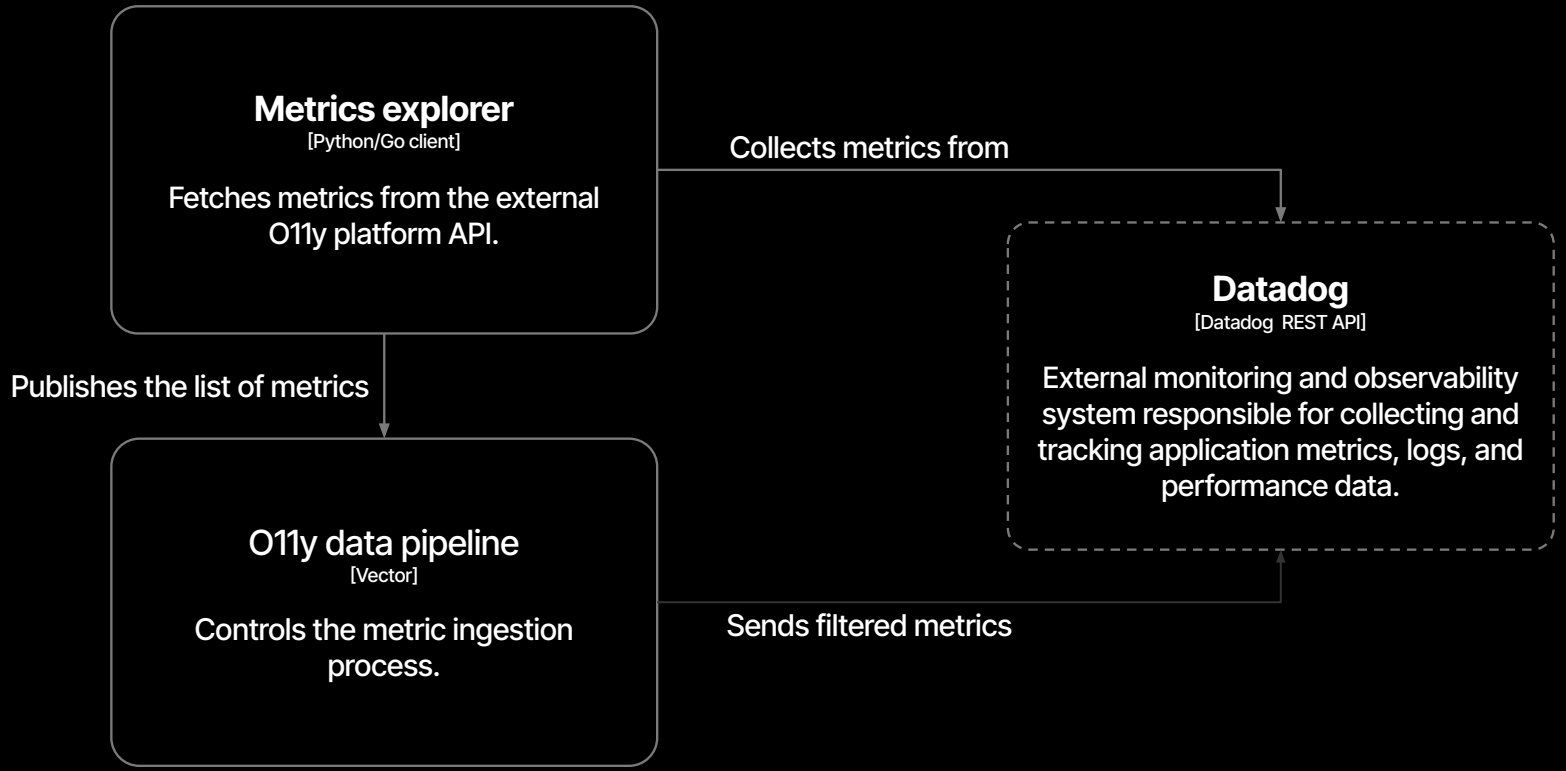
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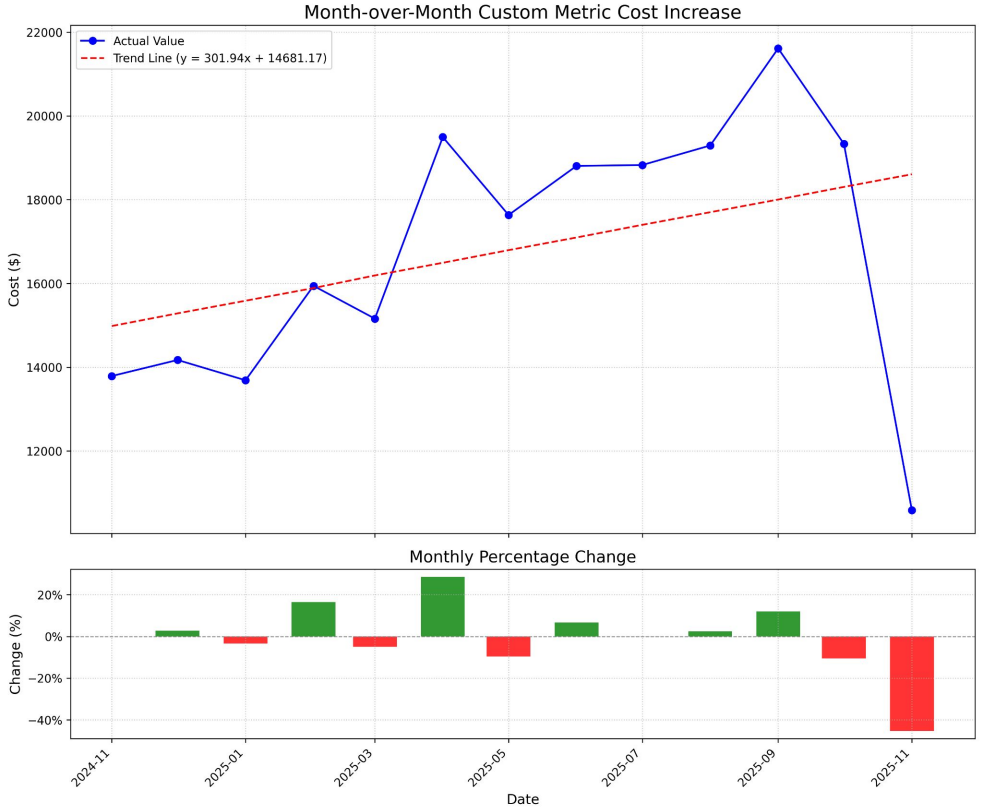


Automation (ACLs generation)



Impact

- Filtering 26.2 TiB out of 68.4 TiB (~38%, daily).
- ~30% cost reduction on custom metrics.
- Cleaner metrics list.



Learnings



Be mindful of your users



Use data & insights to drive your decisions



Retrospectively analyse your team capacity & frustrations



Focus on incremental delivery

QA


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